

First search for cumulative vacuum index creep using cryogenic silicon cavity drift logs: open-data analysis and sensitivity projection

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Abstract

We propose a search for cumulative vacuum refractive index creep, $\Delta n_{\text{vac}}(t) = \alpha_E \int P_{\text{circ}} I_{\text{on}} dt$, using long-term drift logs of cryogenic single-crystal silicon cavities. Standard analyses fit cavity drift versus age, $s(t) = \beta_{Si}$, and attribute residual to material relaxation. We introduce Protocol v3.1: $s_i = \beta_{Si}(t) + \alpha \langle P_{\text{circ}} \rangle_i + \gamma_1 \Delta T_i + \gamma_2 \Delta T_i^2$, with finesse aging $P_{\text{circ}} = P_{\text{trans}} \mathcal{F}/\pi$ and raw frequency (no unwrap needed for published frequency; see Sec. 2).

Pipeline validated on synthetic null and injected $\alpha = 3 \times 10^{-4}$: 30/30 tests pass, null $\hat{\alpha} = -6.4 \pm 4.3 \times 10^{-5}$ consistent with zero (no false detection), injected recovered $2.36 \pm 0.43 \times 10^{-4}$, $R^2 > 0.999$, γ recovers its injected value.

Using only published Fig. 3b from Lee et al., PRL 136, 033801 (2026) [1] for Si2, Si3, Si5 (124K) and Si6 (17K) over > 10 yr, digitized settled drifts are Si2 -57 ± 8 , Si3 -20 ± 4 , Si5 -18 ± 13 , Si6 $+18 \pm 13$ $\mu\text{Hz/s}$, with Si2 integral -29.5 kHz over days 260–3798 vs published -44 kHz/10yr (missing steep days 0–260 imply -646 $\mu\text{Hz/s}$ mean, cf Si3 -883 at day 269). No $P_{\text{trans}}(t)$ is published. We therefore report conditional bound: if the circulating power varied by $\geq X$ across epochs with no correlated drift beyond settled scatter σ , then $|\alpha| \leq \sigma/\Delta P$. For Si6 settled $\sigma = 12.5$ $\mu\text{Hz/s}$ at $P_{\text{circ}} \approx 13.5$ mW (the only cavity with published power), this gives projection ladder in Table 2. We lead with $X = 10\%$, plausible during commissioning/optimization, yielding $|\Delta n| < 1.5 \times 10^{-13}/\text{yr}$ per 100 μW in cryogenic UHV at 17–124K. Ladder is 1.5×10^{-12} (1%), 1.5×10^{-13} (10%), 1.5×10^{-14} (factor 2). Large models requiring $\gtrsim 10^{-12}/\text{yr}$ per 100 μW excluded only if $\geq 1\%$ excursions can be established; reaching 10^{-19} requires measured $P_{\text{trans}}(t), T_{\text{cryo}}(t), \mathcal{F}(t)$ or intentional dither. Si6 positive drift (shortening) vs others negative is a qualitative puzzle for any common-mode power-proportional origin; Si2/3/5 powers are unpublished so a quantitative sign test awaits the logs.

1 Introduction

Vacuum refractive index from strong fields is searched as instantaneous $n_2 I$ with no memory: PVLAS [8], BMV [9], OVAL [10]. Lorentz-violation and α -variation searches using Si cavities compare cavity vs atom and look for sinusoidal residuals, explicitly removing linear drift [12, 13]. Cumulative grooving, where photon dose creates persistent Δn , has not been searched. The Euler–Heisenberg effective action [5] predicts an instantaneous vacuum Kerr index $n_2 \approx 1.6 \times 10^{-33}$ cm^2/W [6] with no cumulative term; PVLAS bounds static-field vacuum birefringence at $\Delta n = (12 \pm 17) \times 10^{-23}$ at 2.5 T [8]. Axion-like particles add instantaneous EHS-like terms [7] and dispersive corrections [6], while domain walls imprint discrete jumps rather than dose accumulation; we know of no concrete model predicting $\Delta n_{\text{vac}} \propto \int P dt$, adopted here as a purely phenomenological parametrization.

Cryogenic Si cavities in cryogenic UHV, operated at 17–124K in this analysis, drift at $\sim 10^{-19}/\text{s}$ fractional [1] — roughly $10^3 \times$ slower than typical ULE cavities (kHz/day vs Hz/day [1]), enabling a DC search.

2 Protocol v3.1

Clock ratios (ν_{Yb+}/ν_{Sr}) are atoms-vs-atoms, servoed to 10^{-18} ; cavity drift is erased from them by construction. The servo’s AOM correction $\nu_{corr}(t)$ is atoms-minus-cavity: the raw DC ruler for cavity length plus vacuum index. With atoms absolute and cavity $\nu = mc/2nL$:

$$\Delta\nu_{corr}(t) = -\nu_0 \left(\frac{\Delta L_{Si}}{L} + \Delta n_{vac} \right) \quad (1)$$

Previous versions fit $s(t) = \beta$. v3.1 fits drift rates $s_i = d\nu_{corr}/dt$ per 10-day epoch, with limitations noted in Sec. 5:

$$s_i = \beta_{Si}(t) + \alpha \langle P_{circ} \rangle_i + \gamma_1 \Delta T_i + \gamma_2 \Delta T_i^2 \quad (2)$$

with:

- $s_i = d\nu/dt$ per 10-day bin from 10-day binned *frequency* vs maser/optical clock, not beat-note phase. PTB/JILA track cavity frequency against masers calibrated to timescales (e.g. UTC(NIST) [4]) or against optical clocks with standard systematic corrections; if re-analyzing raw beat phase, unwrap 2π before differentiating (a 2π slip injects $1/\tau$ Hz steps: ~ 100 mHz at 10 s sampling). For Fig. 3b data [1] we use published frequency drift, so no unwrap needed, but code retains unwrap for raw logs. If labs log absolute AOM frequency (e.g. 80 MHz $\pm$ kHz correction), subtract the nominal first.
- $\langle P_{circ} \rangle_i = \langle P_{trans} \rangle_i \mathcal{F}_i / \pi$, $\mathcal{F}$ from periodic ring-down, interpolated as $\mathcal{F}(t)$ when available; else P_{trans} itself as proxy regressor α_{trans} (conservative: underestimates P_{circ} if $\mathcal{F}$ degraded).
- ΔT_i cryostat temperature, quadratic term for 124K zero-crossing; at 124K, $d\alpha/dT \sim 10^{-11}/K^2$, so 0.1 mK $\rightarrow 10^{-15}$.
- Allan-weighted: $w_i = 1/\sigma_{Allan,i}^2$

α target, $\beta_{Si}(t)$ aging (see Sec. 5 for decay), γ thermal. With constant P , β and $\alpha\langle P \rangle$ degenerate. Variation ΔP breaks degeneracy.

The cumulative (fatigue) form fits $\Delta\nu_{corr}(t) = \beta_{Si}t + \alpha_E E_{cum}(t) + \gamma \int \Delta T dt' + \sum C_i \mathbf{1}_i + \epsilon$ with $E_{cum}(t) = \int P_{circ} I_{on} dt$ never reset (flatlines when unlocked); step dummies C_i absorb relock offsets. Unlock/mode-jump steps are handled by epoch grouping, fitting slopes only within locked epochs.

Pipeline: (1) ingest ν_{corr} , P_{trans} , T , I_{on} ; (2) keep thermal as regressors, never pre-subtracted, preserving covariance; (3) WLS epoch regression plus cumulative cross-check; (4) H_0 : $\alpha = 0$ bound. Implementation: <https://github.com/pleblond/vacuum-grooving-bound>.

3 Pipeline validation (synthetic)

Synthetic s_i with known α_{true} , β , γ , noise matching Si2 settled scatter 8 μ Hz/s.

Case	$\hat{\alpha}$	Result
Null $\alpha = 0$	$-6.4e-05 \pm 4.3e-05$	consistent zero, no false detection
Injected $\alpha = 3e-4$	$2.36e-04 \pm 4.3e-05$	20% low (1.5σ seed noise), $R^2 > 0.999$
Thermal deconfound	γ recovers injected value	validated
Finesse/proxy	$\mathcal{F}$ vs P_{trans} agreement	validated
Fail-safe	30/30 tests pass	

Pipeline does not hide power correlation in β .

4 Open data: Lee et al. 2026 Fig 3b

Digitized Fig. 3b drift rates [1] vs cavity age (time since optical contact bonding).

Cavity	Settled drift [$\mu\text{Hz/s}$]	Span integral
Si2 (21cm,124K,SiO2/Ta2O5)	-57 ± 8	$-29.5 \text{ kHz over days } 260\text{--}3798 \text{ (paper } -44 \text{ kHz/10yr)}$
Si3 (21cm,124K,SiO2/Ta2O5)	-20 ± 4	-27.4 kHz
Si5 (21cm,124K,GaAs/AlGaAs)	-18 ± 13	-20.9 kHz
Si6 (6cm,17K,GaAs/AlGaAs)	$+18 \pm 13$	$+4.6 \text{ kHz}$

Si2 integral check validates digitization. Si6 sign flip: positive (shortening, glass-like) vs others negative (lengthening, Si-like). Only Si6's $P_{\text{circ}} \approx 13.5 \text{ mW}$ is published ($90 \text{ nW} \times 470000/\pi$); Si2/3/5 powers are unknown.

5 α -sensitivity projection

No $P_{\text{trans}}(t)$ published. We report conditional bound: if circulating power varied by $\geq X$ across epochs with no correlated drift beyond settled scatter σ , then $|\alpha| \leq \sigma/\Delta P$.

Table 1: Projection ladder: bound on $|\alpha|$ and $\Delta n/\text{yr}$ per $100 \mu\text{W}$ if power excursions of size X occurred with no observed drift correlation beyond scatter σ . For Si6 $\sigma = 12.5 \mu\text{Hz/s}$, $P_{\text{circ}} \approx 13.5 \text{ mW}$ (90 nW transmission $\times 470000/\pi$). 1% row requires establishing $\geq 1\%$ excursions; lead defensible number is 10% row, plausible during commissioning/optimization when P_{trans} is tuned for RAM. Table is projection, not measurement, until ΔP confirmed from logs.

Excursion X (assumed)	Bound $ \alpha $ [Hz/s/W]	$\Delta n/\text{yr}$ per $100 \mu\text{W}$
1% (0.135 mW)	$9.3e-02$	$1.5e-12$
10% (1.35 mW)	$9.3e-03$	$1.5e-13$
100% (factor 2, 13.5 mW)	$9.3e-04$	$1.5e-14$

Using $\Delta\nu/\nu = -\Delta L/L = -\Delta n(L_{\text{vac}}/L)$, $L_{\text{vac}} \sim L$, $\nu = 194 \text{ THz}$. Lead defensible number is $1.5 \times 10^{-13}/\text{yr}$ per $100 \mu\text{W}$ under $\geq 10\%$ excursion assumption. 1.5×10^{-12} requires establishing only 1% excursions, which is an assumed stability figure, not documented variation.

Target 10^{-19} is ~ 6 orders below 10% projection. Requires measured variation.

6 Further open data before requesting logs

1. Differential Si2-Si3: Matei et al. PRL 118, 263202 (2017) reports $100 \mu\text{Hz/s}$ relative drift [2]. Common-mode maser drift cancels. Power differences are undocumented, but Matei et al. measured the DC photothermal coefficient $1.7(2) \times 10^{-15}$ per μW transmitted for both cavities and run with no active intensity stabilization [2], so multi-% drifts over years are plausible and the logs would settle it. Differential α bound cleaner.
2. Power-dependent drift already seen: Robinson et al. (Si4, 4K) found drift linear in incident power, lowest $-3 \times 10^{-19}/\text{s}$ at 40 nW transmitted and extrapolating to zero at zero power, with DC intensity coupling $1 \times 10^{-12}/\mu\text{W}$ and power stabilized to pW [4]. Any grooving signal must be disentangled from this effect; its mechanism is the key systematic of this search. Crystalline-coating power-dependent systematics (birefringent and photoinduced noise) are further documented [14, 15].

3. Zhang et al. PRL 119, 243601 (2017) bounded 4K drift at < 1 mHz/s (5×10^{-18} /s) over 3 days [3]; long-term 4K drift is unreported, so $\beta_{Si} \approx 0$ at 4K is an extrapolation, not a measurement.
4. Dynamic photothermal response $H^{(PT)}(f)$ was characterized on GW-like dielectric coatings at 1064 nm, room and cryogenic [11], but not on these mirrors: the DC numbers that matter are Matei's $1.7 \times 10^{-15}/\mu\text{W}$ transmitted at 124K [2] and Robinson's $1 \times 10^{-12}/\mu\text{W}$ at 4K [4] — both frequency offsets $\propto P$, distinct from a drift rate $\propto P$. Dither waveforms separate offset from slope.

7 Limitations: $\beta_{Si}(t)$ aging

Fig. 3b shows $1/t$ -ish decay of drift rate with age, visible for Si2 from ~ -200 to -57 $\mu\text{Hz/s}$ over 10yr. Constant- β is limitation of this note.

Cheapest fixes, in order:

- (a) Split record into early/late windows with separate β per window: 2 extra parameters, kills most bias. Works for Si2 settled vs unsettled. Implement as $\beta(t) = \beta_{\text{early}}$ for $t < t_{\text{split}}$, β_{late} else.
- (b) Fit $\beta(t) = \beta_0 + \beta_1/t$ or $\beta_0 + \beta_1 \exp(-t/\tau)$ jointly with α : more principled, matches visible decay. Exponential fitted for Si2 (days ≥ 800): $\tau \approx 1.8$ yr, $\beta_0 = -52$ $\mu\text{Hz/s}$, residuals 11 $\mu\text{Hz/s}$, matching the windowed-late scatter; $1/t$ gives an unphysical -4 asymptote and is disfavored.
- (c) Do nothing here, but caveat: for this note we fit settled window only (last 3 yr) where $\beta(t)$ approximately constant, and note constant- β as limitation. Full analysis will model aging.

For this note we adopt (c): use settled window Si2 -57 ± 8 $\mu\text{Hz/s}$ where aging slope < 5 $\mu\text{Hz/s/yr}$, and state constant- β limitation explicitly. Future work with $P_{\text{trans}}(t)$ logs will adopt (a) or (b). Code supports (a) via `beta_window_edges` and (b) via `fit_exp_aging` (τ grid search + WLS), both validated on synthetic decaying- β logs where single- β flips sign.

8 Outlook to 10^{-19}

- **Free:** Request 10-day mean P_{trans} , T_{cryo} , $\mathcal{F}$ 2015-2026 from PTB (Legero/Sterr) and JILA (Ye). Even single 10% power step during commissioning improves bound to 10^{-13} .
- **2-month dither:** Run Si6 alternating 13.5 mW / 27 mW monthly and synchronously demodulate the slope difference, in which common drift β cancels. With per-month repeatability σ_m (beta wander + thermal; Allan floor $\sim 2 \times 10^{-11}$ Hz/s negligible), $\sigma_\alpha = \sqrt{2}\sigma_m/(\Delta P\sqrt{K})$ over K month-pairs, i.e. $\Delta n/\text{yr} \approx 1.7 \times 10^{-15}(\sigma_m/1 \mu\text{Hz/s})/\sqrt{K}$ per $100\mu\text{W}$: 1.7×10^{-15} in 2 months ($K = 1$), 7.0×10^{-16} in one alternating year ($K = 6$) at optimistic $\sigma_m = 1 \mu\text{Hz/s}$; 5.1×10^{-15} / 2.1×10^{-15} at realistic $\sigma_m = 3 \mu\text{Hz/s}$.
- **Cross-lab:** α_{PTB} vs α_{JILA} at same vacuum, different mounts/coatings. Common α with same sign $\propto P$ would disfavor material creep.

Data and code: <https://github.com/pleblond/vacuum-grooving-bound> (validation notebook, digitization).

References

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